

Modeling

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Practice quiz

You want to hire a group of workers for your job site. You have at your disposal 7 candidates:

- Asmodeus,
- Belphegor,
- Beelzebub,
- Leviathan,
- Lucifer,
- Mammon,
- Satan.

Each decision is modeled by a binary variable x_i which is 1 if the worker i is hired, and 0 otherwise. Model the following constraints:

1. not all candidates can be hired;
2. at least one candidate must be hired;
3. Asmodeus can not be hired if Beelzebub is;
4. Leviathan can be engaged only if Belphegor is;
5. Asmodeus and Lucifer are not separable, and have to be hired both, or not at all;

6. either at least one of the candidates in the group Asmodeus, Belphegor, Beelzebub is chosen, or at least two in the group Belphegor, Leviathan, Lucifer, Mammon.

Provide a feasible solution.